

Yanghao Chen

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Education

Tongji University

Bachelor of Engineering in Engineering Mechanics

GPA: 87/100

Shanghai, China

Sep. 2023 – Present; expected Jun. 2027

Johns Hopkins University

Visiting Undergraduate Research Intern, Department of Civil and Systems Engineering

Baltimore, MD, USA

Jan. 2026 – Oct. 2026

Research Interests: Solid Mechanics, Scientific Machine Learning, Neural Operators.

Research Projects

GPU-Accelerated Hybrid FEM–Neural Operator Solver

Jan. 2026 – Present

Lead Undergraduate Researcher; Advisor: [Prof. Somdatta Goswami](#)

Johns Hopkins University

- Developed a hybrid FEM–neural operator solver based on non-overlapping domain decomposition, using FEM and neural operators in separate subdomains.
- Generated data using GPU-accelerated JAX-FEM simulations with prescribed interface displacements; trained Transolver in JAX to learn mappings from interface displacement fields to subdomain stress fields.
- Computed interface reaction forces from stresses predicted in the neural-operator subdomain; passed them to FEM and fed the updated interface displacements back to the neural operator at each coupling iteration.

PINNs for Shock Capturing in the Burgers Equation

Nov. 2025 – Mar. 2026

Undergraduate Researcher; Advisor: [Prof. Xianyang \(Tom\) Chen](#)

Tongji University

- Trained a PINN in PyTorch for the one-dimensional inviscid Burgers shock problem, using PDE-residual, initial-condition, and boundary-condition losses with automatic differentiation.
- Added an artificial-viscosity term with a fixed coefficient to the PDE residual and trained a second PINN to predict the shock profile.
- Compared standard and artificial-viscosity PINN predictions against the analytical inviscid solution, analyzing shock location, transition width, and errors near the discontinuity.

ConvLSTM Prediction of Concentration and Stress Fields in Battery Materials

Jun. 2025 – Oct. 2025

Undergraduate Researcher; Advisor: [Prof. Ying Zhao](#)

Tongji University

- Generated polycrystalline NMC microstructures and automated MATLAB–COMSOL simulations to produce time-aligned image sequences of lithium concentration and von Mises stress.
- Trained separate conditional ConvLSTM models in PyTorch using past image sequences, grain-orientation maps, and C-rate inputs, with MSE and SSIM losses and scheduled sampling.
- Predicted future concentration and stress image sequences, compared them with COMSOL reference sequences, and analyzed error accumulation during autoregressive prediction.

Domain-Specific LLM Fine-Tuning for Mechanics of Materials

Sep. 2024 – Apr. 2025

Undergraduate Researcher; Advisor: [Prof. Ying Zhao](#)

Tongji University

- Built instruction–response datasets from textbooks and technical literature, covering stress analysis, constitutive laws, and failure theories.
- Fine-tuned Qwen2.5-7B with LoRA on Google Colab for mechanics of materials Q&A and concept explanations.
- Released two instruction datasets and two fine-tuned Qwen2.5-based checkpoints on [Hugging Face](#).

Technical Skills

Scientific Machine Learning

PyTorch, JAX, Neural Operators (Transolver, DeepONet), PINNs, ConvLSTM

Computational Mechanics

Finite Element Methods, FEniCSx, Gmsh, COMSOL Multiphysics, ABAQUS

Scientific Computing & Tools

Python, MATLAB, Linux, Git/GitHub, LiveLink for MATLAB

LLM Fine-Tuning

Qwen2.5, LoRA/PEFT, Instruction Dataset Curation

Languages

Chinese (native), Cantonese, English